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"READ YOU LOUD AND CLEAR!"

by Sunny Tsiao

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"READ YOU LOUD AND CLEAR!"



The Story of NASA's Spaceflight Tracking and Data Network

by Sunny Tsiao

Regardless of how sophisticated it may be, no spacecraft is of any value unless it can be tracked accurately to determine where it is and how it is performing.

At the height of the space race, 6,000 men and women operated NASA's Spaceflight Tracking and Data Network at some two dozen locations across five continents. This network, known as the STDN, began its operation by tracking Sputnik 1, the world's first artificial satellite that was launched into space by the former Soviet Union. Over the next 40 years, the network was destined to play a crucial role on every near-Earth space mission that NASA flew. Whether it was receiving the first television images from space, tracking Apollo astronauts to the Moon and back, or data acquiring for Earth science, the STDN was that intricate network behind the scenes making the missions possible. Some called it the "Invisible Network," a hallmark of which was that no NASA mission has ever been compromised due to a network failure.

Read You Loud and Clear! is a historical account of the STDN, starting with its formation in the late 1950s to what it is today in the first decade of the twenty-first century. It traces the roots of the tracking network from its beginnings at the White Sands Missile Range in New Mexico to the Tracking and Data Relay Satellite System (TDRSS) space-based constellation of today. The story spans the early days of satellite tracking using the Minitrack Network, through the expansion of the Satellite Tracking And Data Acquisition Network (STADAN) and the Manned Space Flight Network (MSFN), and finally, to the Space and Ground Networks of today.

Written from a nontechnical perspective, the author has translated a highly technical subject into historical accounts told within the framework of the U.S. space program. These accounts tell how international goodwill and foreign cooperation were crucial to the operation of the network and why the space agency chose to build the STDN the way it did. More than anything else, the story of NASA's STDN is about the "unsung heroes of the space program."